ALLIANCE KCSE 2025 PREDICTION

POSSIBLE KCSE EXAM

PHYSICS

232/2

PAPER 2

(THEORY)

TIME: 2 HOURS

NAME………………………………………………………………………………………

SCHOOL…………………………………………………… SIGN………………………

INDEX NO…………………………………………… ADM NO……………………

Kenya Certificate of Secondary Education.

Instructions to candidates

(a) *Write your name, index number in the spaces provided above.*
(b) *Sign and write the date of examination in the spaces provided above.*
(c) *This paper consists of* **two** *sections:* **A** *and* **B**.
(d) *Answer* **all** *the questions in sections* **A** *and* **B** *in the spaces provided.*
(e) ***All*** *working* **must** *be clearly shown.*
(f) *Silent non-programmable electronic calculators may be used.*
(g) *Candidates should answer the questions in English.*

For Examiners Use Only

Section	Question	Candidate's Score
A	1 – 12	
B	13	
	14	
	15	
	16	
	17	
	Total	80

SECTION A (25 MKS):

Answer all questions in this section in the spaces provided.

1. **Figure 1** shows a ray of light incident on a mirror at an angle of 45^0. Another mirror is placed at an angle of 45^0 to the first one as shown

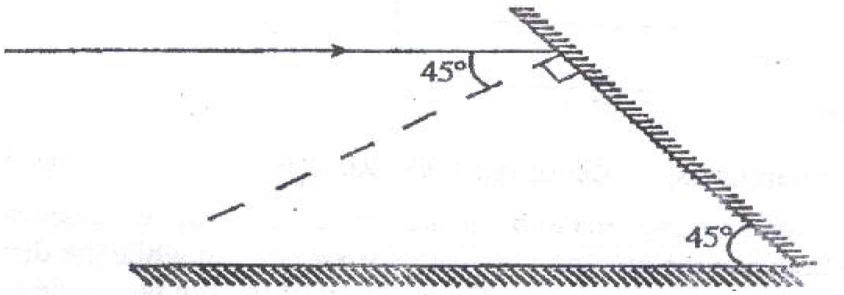


Figure 1

Sketch the path of the ray until it emerges **(2 mks)**

2. **Figure 2** shows a soft iron bar AB placed in a coil near a freely suspended magnet.

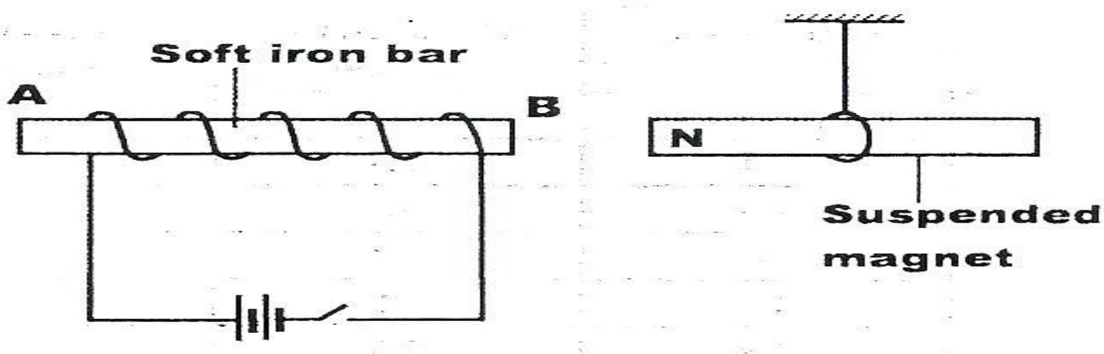


Figure 2

Explain the observation made when the switch is closed. **(2 mks)**

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

3. Table 1 shows radiations and the irrespective frequencies.

Type of radiation	Yellow light	Gamma rays	Radio waves	Micro waves

Frequency (Hz)	1×10^{15}	1×10^{22}	1×10^{6}	1×10^{11}

Arrange the radiations in the order of increasing energy. **(l mk)**

....

....

4. State the reason why electrical power is transmitted over long distances at very high voltages. **(1 mk)**

....

....

5. A boy standing in front of a cliff blows a whistle and hears the echo after 0.5s. He then moves 17 metres further away from the cliff and blows the whistle again. He now hears the echo after 0.6s. Determine the speed of the sound. **(4mks)**

....

....

....

....

....

....

6. **Figure 3** shows a human eye with a certain defect

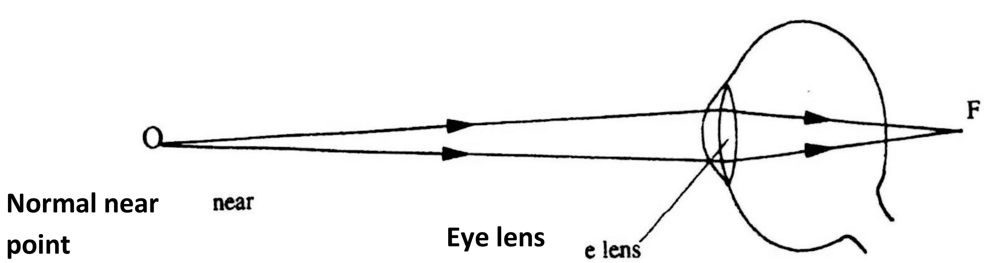


Figure 3 e 1

(i) Name the defect **(1 mk)**

……………………………………………………………………………………………………

ii) On the same diagram, sketch the appropriate Len and rays to show how the defect can be corrected. **(2mks)**

7. Polarisation is a defect of a simple cell. State how it reduces the current produced and how this defect can be minimized **(2mks)**

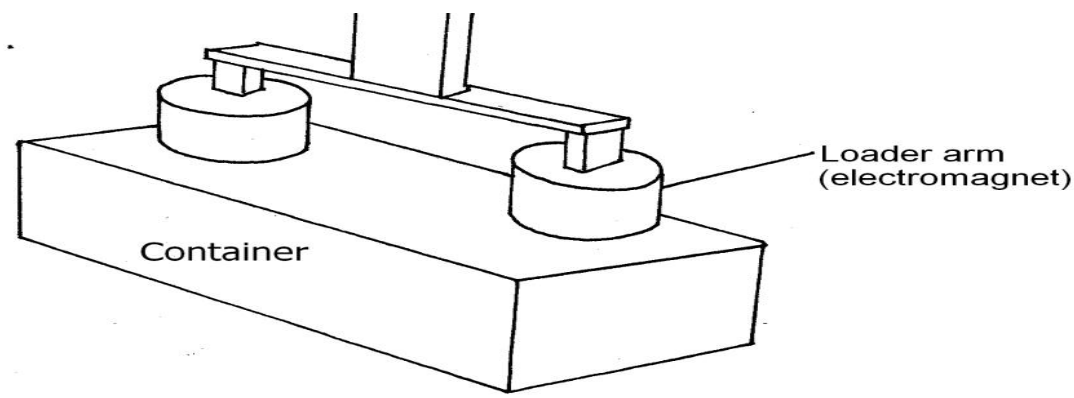
……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

8. The figure below shows container loader which uses electromagnet to offload containers from a ship.



(i) Why should the container be made of iron or steel **(1mk)**

……………………………………………………………………………………………………

……………………………………………………………………………………………………

(ii) State two ways in which the loader can be made to lift heavier container **(2mks)**

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

9. Two 12V lead acid accumulators are rated 60Ah and 70Ah. State two physical differences between the accumulators

(2mks)

....

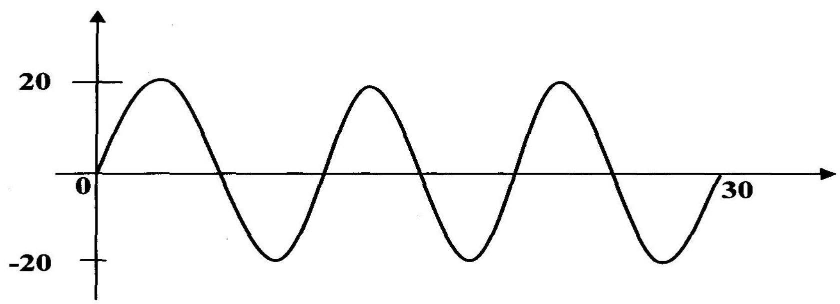
....

....

....

10. The diagram below shows part of a wave form. The numbers on the diagram show scales in meters. If the speed of the wave is $20ms^{-1}$, determine the frequency and wavelength of the wave.

(3mks)



....

....

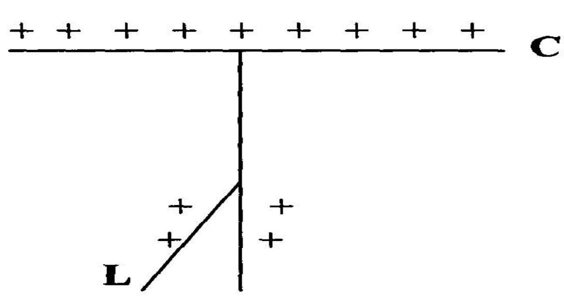
....

....

....

....

11. A gold leaf electroscope is positively charged as shown in the diagram below where **C** is the cap and **L** is the gold leaf. State and explain what happens to **L** when a positively charged rod is brought near **C** without touching it. **(2mks)**



....

……………………………………………………………………………………………………

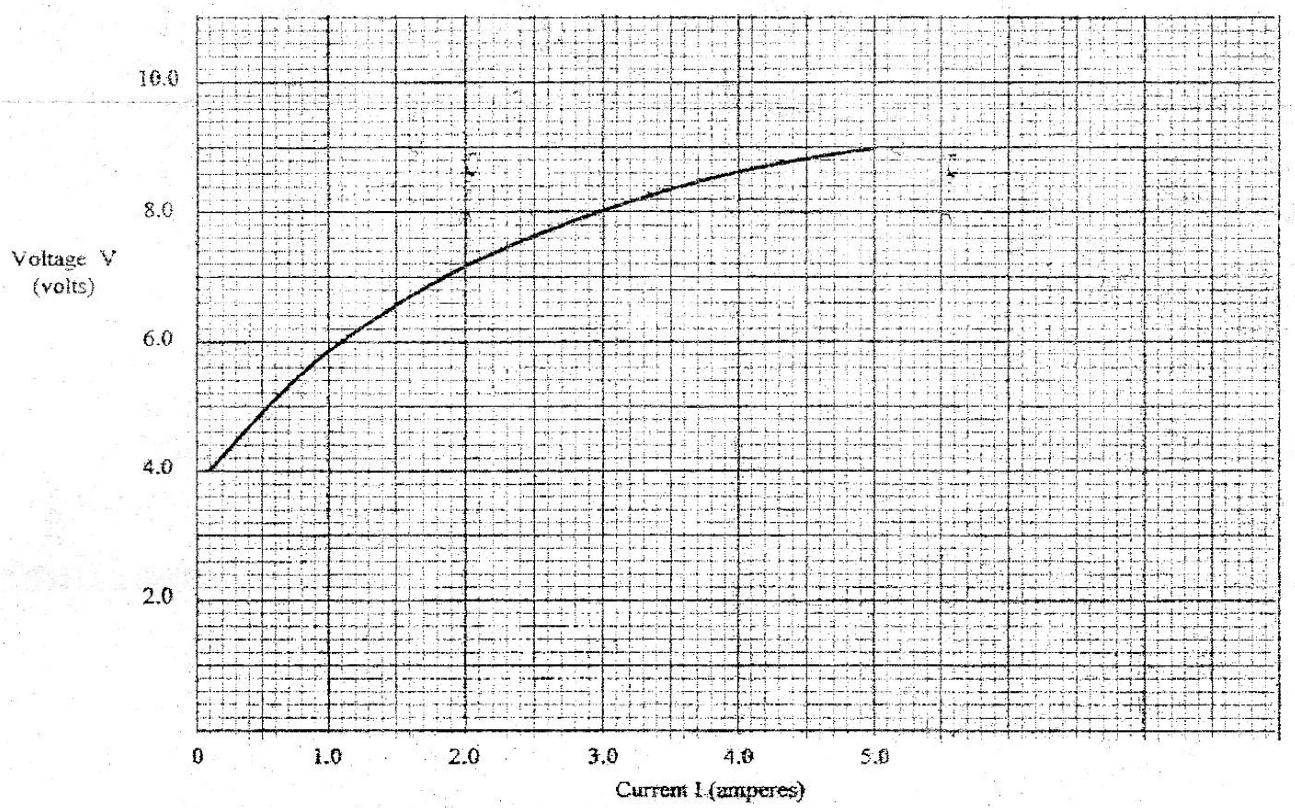
……………………………………………………………………………………………………

……………………………………………………………………………………………………

SECTION B (55 Marks)

Answer all the questions in this section in the spaces provided

12 a) Figure 8 shows graph of potential difference V (volts) against a current (ampere) for a certain device.



From the graph

(i) State with a reason whether or not the device obeys ohms law **(2marks)**

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

(ii) Determine the resistance of the device at

(I) 1 = 1.5A

(II) 1 = 3.5A

……

……

……

……

(iii) From the results obtained in (ii) translate how the resistance of the device varies as the current increases **(1mark)**

……

……

……

(iv) State the cause of this variation in resistance **(1mark)**

……

……

b) Three identical dry cells each of e.m.f. 1.6V are connected in series to a resistor of 11.4Ω. A current of 0.32A. How is the circuit? Determine:

(i) The total e.m.f. of the cell **(1mark)**

……

……

……

(ii) The internal resistance of each cell **(3marks)**

……

……

……

……

13 (a) State the meaning of the term ‘principal focus’ as applied in lenses **(1mark)**

……

……

(b) You are provided with the following apparatus to determine the focal length of a lens:

- A biconcave lens and lens holder
- A lit candle
- A white screen
- A metre rule

(i) Draw a diagram to show how you would arrange the above apparatus to determine the focal length of the lens **(1mark)**

(ii) Describe the procedure you would follow

………

………

………

(iii) State two measurements that you would take **(2marks)**

………

………

(iv) Explain how the measurements in (iii) would be used to determine the focal length.

………

………

(c) An object is placed 30cm in front of a concave lens of focal length 20cm. Determine the magnification of the image produced. **(4marks)**

………

………

………

………………………………………………………………………………………………………

………………………………………………………………………………………………………

14 (a) State what is meant by 'electromagnetic induction' **(1mark)**

………………………………………………………………………………………………………

………………………………………………………………………………………………………

(b) Figure 9 shows a simple electric generator

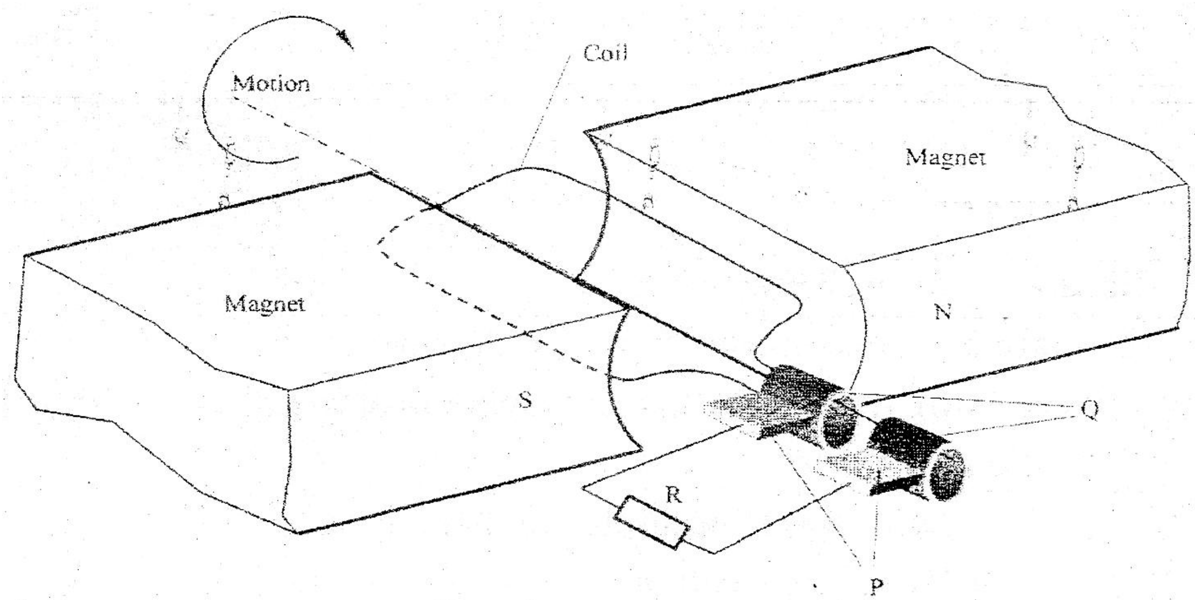


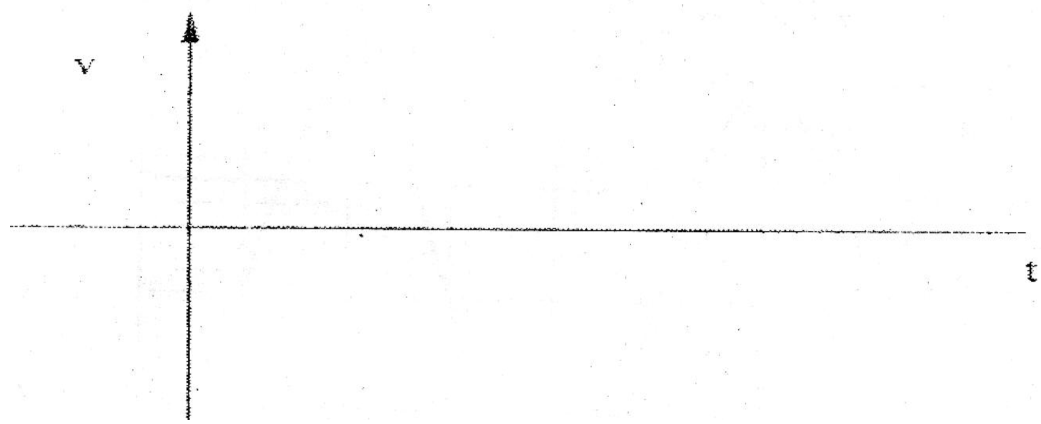
Figure 9

(i) Name the parts labeled P and Q **(2marks)**

………………………………………………………………………………………………………

………………………………………………………………………………………………………

(ii) Sketch on the axes provided a graph to show how the magnitude of the potential difference across R changes with the line I



(iii) State two ways in which the potential differences produced by such a generator can be increased. **(2marks)**

....

....

....

(c) In a transformer, the ratio of primary to the secondary turns is 1:10. A current of 500mA flows through a 200Ω resistor in the secondary circuit. Assuming that the transformer is 100% efficient; determine;

(i) The secondary voltage **(1mark)**

....

....

(ii) The primary voltage **(2marks)**

....

....

....

....

(iii) The primary current **(2marks)**

....

....

....

....

15 **(a)** State two differences between cathode rays and electromagnetic radiations

....

....

....

(b) Figure 10 shows the main features of a cathode ray oscilloscope (CRO)

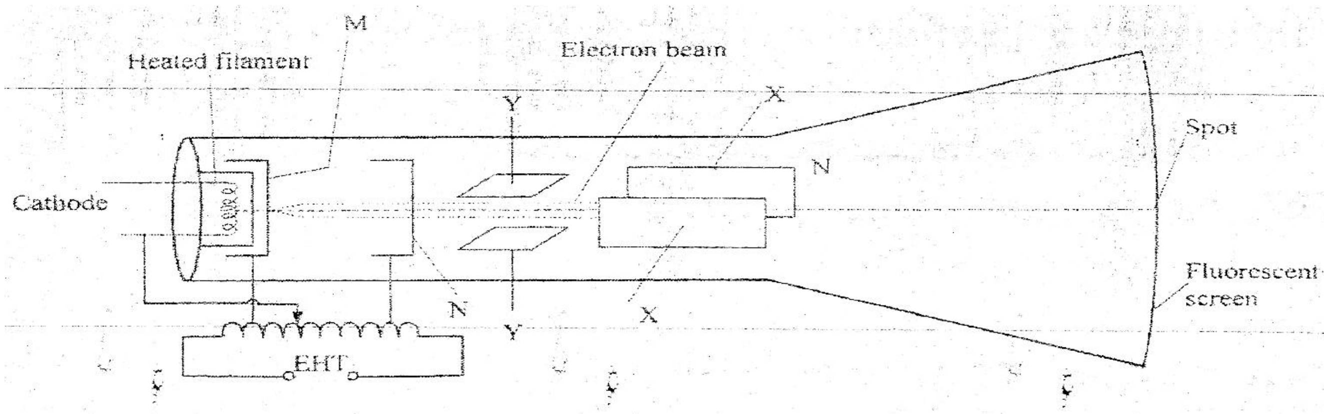


Figure 10

(i) Name the parts labeled M and N

....

....

(ii) Explain how electrons are produced in the tube.

....

....

(iii) When using the CRO to display waveforms of voltages state where the following should be connected

(I) The voltage to be displayed on the screen

....

(II) The time base volume

....

(iv) State why the tube is highly evacuated

....

....

(c) Figure 11 shows the waveform of a voltage displayed on the screen of a CRO. The Y gain calibration was 5V per cm

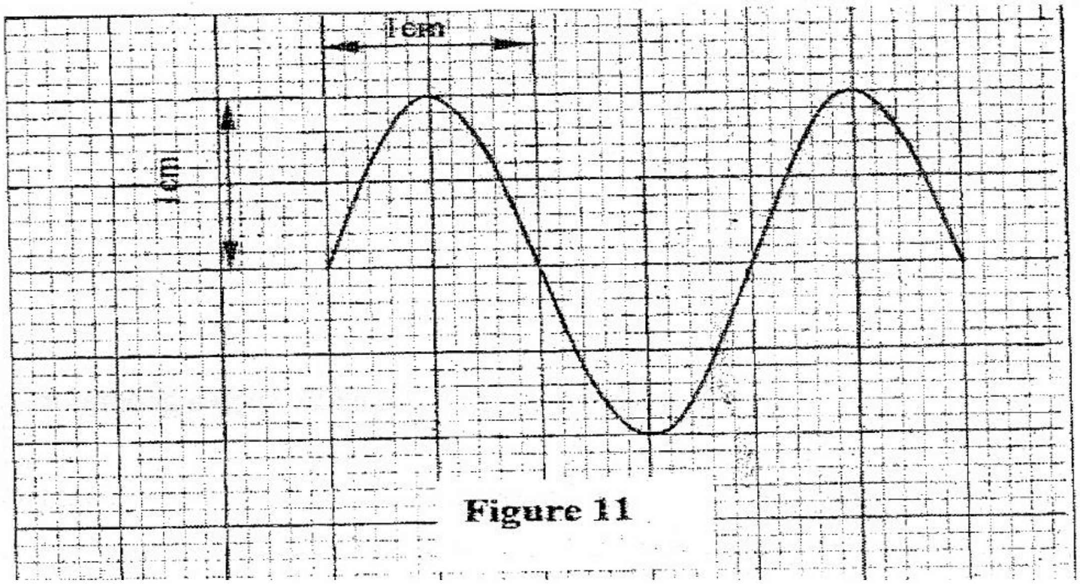


Figure 11

(i) Determine the peak-to-peak voltage of the Y input

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

(ii) Sketch on the same figure the appearance of the waveform after the voltage of the input signal is halved and it's frequency is doubled **(2marks)**

16. a) It is observed that when ultra – violet radiation is directed onto a clean Zinc plate connected to the cap of a negatively charged leaf electroscope, the leaf falls.

i. Explain this observation. **(1mark)**

……

……

ii. Explain why the leaf of the electroscope does not fall when infrared radiation is directed onto the zinc plate. **(1mark)**

……

……

b) State the effect on the electrons emitted by the photoelectric effect when the intensity of incident radiation is increased. **(1mark)**

……

……

c) The maximum wavelength required to cause photoelectric emission on a metal surface is 8.0×10^{-7} m. The metal surface is irradiated with light of frequency 8.5×10^{14} Hz.

(**Take lev = 1.6×10^{-19} J , c = 3.0×10^{8} m/s , h= 6.63×10^{-34} Js)**

Determine:

i). The threshold frequency. **(2marks)**

……

……

……

……

ii) .The work function of the metals in electron volts. **(2marks)**

……

……

………

………

iii) . The maximum Kinetic energy of the electrons. **(3marks)**

………

………

………

………

………

THIS IS THE LAST PRINTED PAGE